

Practical # 02

Objective:

Develop programs to perform insertion, deletion, and traversal operations on a one-dimensional array.

Theory:

A one-dimensional array is a structured collection of elements of the same data type, stored in contiguous memory locations. Each element can be accessed using an index. Arrays are widely used for storing and processing homogeneous data efficiently.

In this lab, we will explore how to:

- Declare and initialize a one-dimensional array.
- Perform **traversal** operations to access array elements.
- Perform **insertion** operations (at beginning, end, or specific position).
- Perform **deletion** operations (from beginning, end, or specific position).

Syntax of a one-dimensional array:

Data_Type Array_Name[Int_Value];

Example program: *Declare and print elements of one-dimensional Array.*

```
#include<iostream>
using namespace std;
int main()
{    /*an array with 5 rows*/
    int a[5]= {10,20,30,40,50};
    int i;
    /* Output each array elements value*/
    for(i=0; i<5; i++)
    {
        cout<<"element"<<i<<"="<<a[i]<<"\n";
    }
}
```

OUTPUT

```
element0=10  
element1=20  
element2=30  
element3=40  
element4=50
```

Review Questions/ Exercise:

1. **Develop** a C++ program to find and display the largest value in a one-dimensional array.
2. **Implement** insertion of a new item at the beginning of an array and traverse to display all elements.
3. **Implement** insertion of a new item at the end of an array and traverse to display all elements.
4. **Develop** a program to insert a new item at a user-specified location in the array and traverse to display all elements.
5. **Implement** deletion of an item from the beginning of an array and traverse to display the updated array.
6. **Implement** deletion of an item from the end of an array and traverse to display the updated array.
7. **Develop** a program to delete an item from a user-specified location in the array and traverse to display the updated array.
8. **Construct** a program to search for an element in an array using either index or value.
9. **Develop** a program to update the value of an element at a given index.
10. **Implement** a program to add two integer arrays to produce and display the output array.

Name: _____

Roll #: _____

Date: _____

Subject Teacher